COFFEE SHOP SALES DATA ANALYSIS

USING EXCEL

Transforming Raw Sales Data into Smarter Business Decisions

Tool
Microsoft Excel

Skills
Data Cleaning • EDA
Pivot Tables • Forecasting

Outcome
Actionable Business Insights

Report created by: R.Dunith Nadupa | Date: 14.04.2026

Coffee Shop Sales Data Analysis using Excel

Introduction

- In today's data-driven world, businesses rely heavily on data analytics to make informed and strategic decisions. This project focuses on analyzing a real-world **coffee shop sales dataset from the United States**, with the objective of transforming raw transactional data into meaningful insights that can drive business growth.

- The dataset consists of over **149,000 sales transactions**, capturing detailed information such as product categories, transaction timestamps, store locations, and revenue. However, raw data alone does not provide value unless it is properly cleaned, structured, and analyzed.

- Through this project, I applied key data analytics techniques using **Microsoft Excel** to simulate a real business scenario. The process began with **data cleaning and preparation**, ensuring accuracy and consistency, followed by **Exploratory Data Analysis (EDA)** to identify trends and patterns in customer behavior.

- By leveraging **Pivot Tables and data visualization techniques**, I was able to uncover critical business insights such as best-selling products, peak sales periods, and store performance comparisons. Furthermore, I utilized Excel's **forecasting capabilities** to predict future sales trends, demonstrating how historical data can be used for strategic planning.

- This project highlights how even a widely accessible tool like Excel can be used to perform powerful data analysis and support data-driven decision-making in a retail business environment. It also reflects the practical role of a data analyst in converting raw data into actionable insights that create real business value.

- The analysis focuses on understanding:

    - Customer purchasing behavior
    - Product performance
    - Sales trends over time

Objectives

i. **Identify best-selling products/categories →** which drinks or bakery items bring the most sales volume/revenue?

ii. **Analyze peak sales hours & days →** When are customers most active (time of day, day of week)?

iii. **Compare store locations →** which branch (store_location) performs best, and what products drive that?

iv. **Estimate future sales →** Use simple forecasting to project next month's sales.

Dataset Description

- Raw Dataset Overview:

  Total Records: **149,116 transactions**
  Columns: **11 variables**

- Key Variables:
  - Transaction ID
  - Transaction Date & Time
  - Product Category & Type
  - Product Detail
  - Store Location
  - Sales Amount
  - Unit Price
  - Transaction Quantity

Feature Engineering:

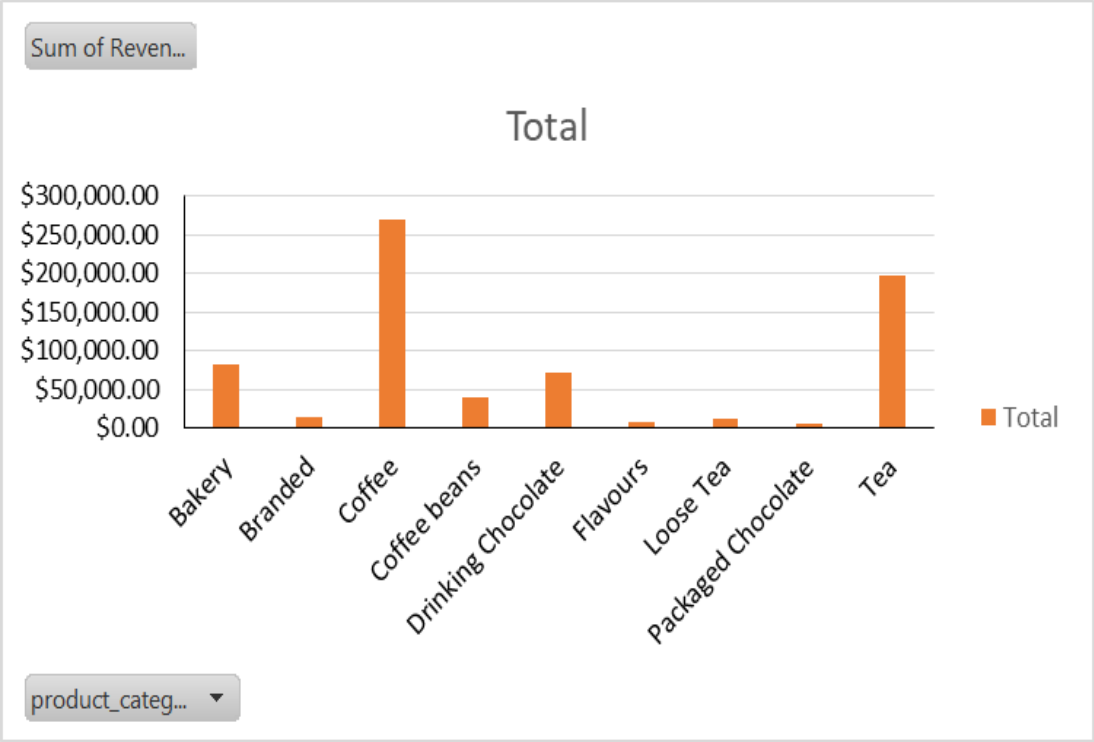
- Created new variables to enhance analysis:
  - **Hour of Day** → to identify peak hours
  - **Day of Week** → to analyze weekly trends
  - **Sales Aggregation** → daily revenue

Exploratory Data Analysis (EDA)

1) Best-Selling Products

- Which drinks or bakery items bring the most sales volume/revenue?

Row Labels	Sum of Revenue
Bakery	$82,315.64
Branded	$13,607.00
Coffee	$269,952.45
Coffee beans	$40,085.25
Drinking Chocolate	$72,416.00
Flavours	$8,408.80
Loose Tea	$11,213.60
Packaged Chocolate	$4,407.64
Tea	$196,405.95
Grand Total	**$698,812.33**



- Coffee and beverage items dominate sales.
- Certain product sizes (e.g., Large) generate higher revenue.

2) Peak Sales Hours & Days

- When are customers most active (time of day, day of week)?

(Days)

Sum of Revenue	Column Labels							
Row Labels	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Grand Total
Bakery	$11,537.45	$12,244.17	$11,542.43	$11,863.82	$11,900.17	$11,902.50	$11,325.10	$82,315.64
Branded	$2,093.00	$1,662.00	$2,054.00	$2,209.00	$1,650.00	$1,991.00	$1,948.00	$13,607.00
Coffee	$38,136.55	$39,322.10	$38,376.50	$38,386.20	$38,993.75	$39,568.25	$37,169.10	$269,952.45
Coffee beans	$5,421.95	$5,923.00	$6,182.55	$6,038.45	$4,691.80	$5,704.35	$6,123.15	$40,085.25
Drinking Chocolate	$10,107.25	$10,809.50	$10,084.50	$10,217.50	$10,941.75	$10,152.50	$10,103.00	$72,416.00
Flavours	$1,186.40	$1,155.20	$1,292.00	$1,221.60	$1,074.40	$1,263.20	$1,216.00	$8,408.80
Loose Tea	$1,642.35	$1,491.60	$1,644.90	$1,650.45	$1,620.35	$1,685.85	$1,478.10	$11,213.60
Packaged Chocolate	$743.51	$543.01	$634.06	$626.87	$723.11	$593.40	$543.68	$4,407.64
Tea	$27,461.85	$28,526.70	$27,645.00	$28,099.65	$29,172.45	$28,511.95	$26,988.35	$196,405.95
Grand Total	**$98,330.31**	**$101,677.28**	**$99,455.94**	**$100,313.54**	**$100,767.78**	**$101,373.00**	**$96,894.48**	**$698,812.33**

(Time)

Sum of Revenue	Column Labels			
Row Labels	Afternoon	Evening	Morning	Grand Total
Bakery	$30,839.25	$3,313.69	$48,162.70	$82,315.64
Branded	$4,359.00	$678.00	$8,570.00	$13,607.00
Coffee	$113,876.65	$11,604.00	$144,471.80	$269,952.45
Coffee beans	$10,209.50	$1,697.20	$28,178.55	$40,085.25
Drinking Chocolate	$30,414.75	$3,845.75	$38,155.50	$72,416.00
Flavours	$2,364.80	$267.20	$5,776.80	$8,408.80
Loose Tea	$3,336.35	$491.35	$7,385.90	$11,213.60
Packaged Chocolate	$1,203.64	$227.03	$2,976.97	$4,407.64
Tea	$82,537.40	$9,258.10	$104,610.45	$196,405.95
Grand Total	**$279,141.34**	**$31,382.32**	**$388,288.67**	**$698,812.33**

- Sales peak during:
  - Morning hours (commute time)
  - Specific weekdays (likely workdays)

3) Store Location Performance

- Which branch (store_location) performs best, and what products drive that?

Row Labels	Sum of Revenue
Coffee	**$269,952.45**
Hell's Kitchen	$91,222.65
Astoria	$89,744.30
Lower Manhattan	$88,985.50
Tea	**$196,405.95**
Astoria	$67,839.90
Hell's Kitchen	$64,701.30
Lower Manhattan	$63,864.75
Bakery	**$82,315.64**
Lower Manhattan	$28,328.94
Hell's Kitchen	$27,386.95
Astoria	$26,599.75
Drinking Chocolate	**$72,416.00**
Astoria	$26,335.25
Hell's Kitchen	$23,586.25
Lower Manhattan	$22,494.50
Coffee beans	**$40,085.25**
Hell's Kitchen	$18,635.10
Lower Manhattan	$11,230.95
Astoria	$10,219.20
Branded	**$13,607.00**
Lower Manhattan	$6,208.00
Astoria	$5,457.00
Hell's Kitchen	$1,942.00
Loose Tea	**$11,213.60**
Hell's Kitchen	$4,461.35
Lower Manhattan	$3,558.25
Astoria	$3,194.00
Flavours	**$8,408.80**
Lower Manhattan	$3,767.20
Hell's Kitchen	$2,876.80
Astoria	$1,764.80
Packaged Chocolate	**$4,407.64**
Hell's Kitchen	$1,698.77
Lower Manhattan	$1,619.16
Astoria	$1,089.71
Grand Total	**$698,812.33**

- **High-performing stores →** expansion opportunities
- **Low-performing stores →** improvement strategies

4) Forecasting Analysis

- Used Excel's Forecast tool to predict future sales based on time-series data.

$Y_t = \alpha x_t + (1-\alpha) y_{t-1}$

Forecast Statistics:-

Statistic	Value
Alpha	0.10
Beta	0.00
Gamma	0.90
MASE	1.19
SMAPE	0.06
MAE	305.43
RMSE	397.57

- Sales show **stable patterns with slight fluctuations**
- Forecast indicates expected future demand
- SMAPE ≈ **6% error** → Good prediction accuracy
- MASE ≈ **1.18** → Reliable forecasting model.

Key Findings & Recommendations

- Coffee products generate the highest revenue.
- Peak hours are concentrated in morning periods.
- Sales trends are predictable using historical data.
- Certain store locations outperform others.

Recommendations:

- Increase stock for high-demand products.
- Schedule more staff during peak hours.
- Promote low-performing locations.
- Use forecasting for monthly planning.

Skills Demonstrated:

- Data Cleaning & Preprocessing
- Exploratory Data Analysis (EDA)
- Pivot Tables
- Time Series Forecasting
- Business Insight Generation.

Thank You.